

Mastery Workbook 4

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I have neither given nor received unauthorized assistance.

#1

- a. Calculate $7^3 \bmod 5$ by calculating 7^3 and then using a calculator to find the value using that result.

When using the calculator, $7^3 = 343$. And then when applying $343 \bmod 5 = 3$

- b. What is $7 \bmod 5$?

$$7 \bmod 5 = 2$$

- c. Rewrite $7^3 \bmod 5$ as $(7 \cdot 7 \cdot 7) \bmod 5$, now use the rule from page 242 and rewrite this to use what you found in (b) to find the answer. Show all steps.

According to corollary #2 on page 242 of Rosen:

Let m be a positive integer and let a and b be integers, then:

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

$$(a \cdot b) \bmod m = ((a \bmod m) \cdot (b \bmod m)) \bmod m$$

So, starting by breaking $7^3 \bmod 5$ into $(7 \cdot 7 \cdot 7) \bmod 5$, we have

$(7 \cdot 7 \cdot 7) \bmod 5 = 7^2 \cdot 7 \bmod 5$	rewriting as a product
$= ((7^2 \bmod 5)(7 \bmod 5)) \bmod 5$	using Corollary 2
$= (((7 \bmod 5)(7 \bmod 5) \bmod 5)(7 \bmod 5)) \bmod 5$	using Corollary 2 and substitution
$= (((2)(2) \bmod 5)(2)) \bmod 5$	definition of a modulo and substitution ($7 \bmod 5 = 2$)
$= ((4 \bmod 5)(2)) \bmod 5$	simplifying using multiplication
$= ((4)(2)) \bmod 5$	simplifying using modulo definition
$= 8 \bmod 5$	by multiplication
$= 3$	by definition of a modulo

- d. Find your own simple example of breaking up a large number and using mod rules to calculate “mod” without using a calculator. Work out your example on this page. Post your answer on Piazza too!

For this exercise, let's simplify the expression $1000 \bmod 3$. We know (by definition but also from scientific notation) that $1000 = 10^3$. So from here:

$1000 \bmod 3 = 10^3 \bmod 3$	starting by substituting $1000 = 10^3$
$= 10^2 \cdot 10 \bmod 3$	commutativity under multiplication
$= ((10^2 \bmod 3)(10 \bmod 3)) \bmod 3$	using Corollary 2
$= (((10 \bmod 3)(10 \bmod 3) \bmod 3)(10 \bmod 3)) \bmod 3$	using Corollary 2 and substitution
$= (((1)(1) \bmod 3)(1)) \bmod 3$	definition of a modulo and substitution ($10 \bmod 3 = 1$)
$= ((1 \bmod 3)) \bmod 3$	simplifying using multiplication and fact
$= 1 \bmod 3$	simplifying using modulo definition

and since $1 \bmod 3 = 1$, we have determined $1000 \bmod 3 = 1$.

2. For all positive real numbers a and b is $(\log_{10} a)^b = b \log_{10} a$.

We will **disprove** this by counterexample. To do this, let us pick an $a = 10$ and $b = 2$, then we have

$$2 \cdot \log_{10}(10) = 2 \cdot 1 = 2 \quad \text{by substitution and logarithmic arithmetic } (\log_{10}(10) = 1)$$

and

$$(\log_{10}(10))^2 = 1^2 = 1 \quad \text{by substitution and logarithmic arithmetic } (\log_{10}(10) = 1)$$

and we know that $2 \neq 1$, therefore we have shown by the counterexample where $a = 10$ and $b = 2$, $(\log_{10} a)^b \neq b \log_{10} a$. Thus, $(\log_{10} a)^b \neq b \log_{10} a$ for all positive real numbers a and b .